

Haluk Yuzukirmizi

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EDUCATION

University of Massachusetts Amherst

Expected Graduation : May 2027

B.S. in Computer Science | Business Minor | Commonwealth Honors College

GPA: 3.94

- **Relevant Coursework:** Object Oriented Programming, Data Structures, Programming Methodology, Multivariate Calculus, Discrete Math, Statistics, Algorithms, Linear Algebra, Computer Systems
- **Awards:** Adobe Computer Science Scholarship, Massachusetts High Demand Scholarship, Dean's List (All Semesters)

EXPERIENCE

Commonwealth of Massachusetts - EOTSS

Apr 2025 - Aug 2025

Artificial Intelligence Intern

Amherst, MA

- Engineered a multi-modal Retrieval-Augmented Generation (RAG) chatbot for the Unity High-Performance Computing platform and integrated it to team communication platform (Slack), enabling 24/7 automated support **for 2,000+ researchers**.
- Designed fully serverless Amazon Web Services (AWS) architecture (Bedrock, Lambda, S3) with LangChain integration.
- Built a knowledge base by scraping and auto-syncing 160+ documentation sources with BeautifulSoup, and generated Q&A pairs from Slack conversations to expand the knowledge coverage by **30%** over the documentation.
- Integrated CI/CD pipeline using GitHub Actions and AWS CloudFormation for flexible ECS or API Gateway deployment with **99.99% uptime**.
- Reduced computing costs by **12.3×** and average response time from **14+ to 7.5 seconds** while maintaining **95% model accuracy** through contextual chunking, metadata filtering, and prompt engineering.

University of Massachusetts Amherst - Computer Vision Lab

Sept 2024 - May 2025

Undergraduate Research Associate

Amherst, MA

- Conducted research under co-director Prof. Subhransu Maji on DISCount, a human-in-the-loop machine learning model for object counting in high-resolution images.
- Built modular Python pipeline (OpenCV, NumPy, Pandas, Matplotlib, SciPy) to preprocess images, compute density maps, and run 100+ simulations comparing uniform, triangle, and model-guided sampling.
- Applied importance sampling to minimize labeling effort, achieving **90%+ accuracy with only 20% labeled data** surpassing triangle and uniform distribution baselines and cutting labeling costs by **4×**.

PROJECTS

hyuzukirmizi.github.io/home

DoorSmash (HackUMass XIII, 36 hours) | TypeScript(React), Python(FastAPI), Supabase(PostgreSQL)

[DoorSmash](#)

- Launched a full-stack delivery and nutrition platform serving 100+ users, featuring real-time tracking.
- Developed an agentic AI chatbot (Pydantic AI, Gemini, ElevenLabs) with 20+ specialized tools for personalized nutrition insights, conversational ordering and health recommendations.
- Built an automated web scraper on AWS Lambda that processed 700+ menu items weekly, populated a Supabase backend, and deployed the architecture via 15+ RESTful API endpoints.

S&P 500 Predictive Modeling | Python, Scikit-learn, Pandas, Matplotlib, XGBoost

- Engineered supervised ML pipeline, boosting precision by 44% through rolling-window backtesting and feature scaling.

SKILLS

Programming Languages: Python, JavaScript/TypeScript, C, Java, Rust, C++, Go, Assembly

Frameworks & Libraries: Node.js, Next.js, LangChain, Scikit-learn, Pandas, Numpy, Matplotlib, React

Technologies: AWS, Git, HTML, CSS, Linux, Azure, Jupyter, REST API, Docker, PostgreSQL, MySQL, Agile, Scrum

LEADERSHIP

MIT Gen AI Global for AI 2027 - Model Architecture Team Member

Dec 2025 - Present

- Defined model and agent system boundaries (I/O, tools, rate limits, permissions) to support safety and architecture reviews.
- Analyzed misuse surfaces and interpretability gaps, pairing each with concrete mitigations and review cadence.